



OPTIMAL TRANSPORT IN LINEAR INDEPENDENT COMPONENT ANALYSIS

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Chapter 1

Introduction

1.1 Motivation: Disentangling Latent Variables

Blind Source Separation, specifically formulated as Independent Component Analysis (ICA), is a fundamental problem in modern statistics (Jutten & Herault, 1985). It has widespread applications across diverse fields, including machine learning, signal processing, econometrics, and neuroscience. The primary objective of ICA is to recover hidden, statistically independent sources from an observed set of mixtures. In this thesis, we focus on the linear mixture setting. ICA is frequently contrasted with Principal Component Analysis (PCA); however, while PCA merely ensures that the resulting components are uncorrelated, ICA enforces the much stronger condition of full statistical independence (Hyvärinen, Karhunen, & Oja, 2001). This rigorous criterion allows ICA to successfully disentangle complex latent variables that PCA cannot fully separate.

1.2 Intuition from Central Limit Theorem (CLT)

For the linear ICA model to be identifiable, a strict condition must be met: at most one of the underlying independent sources can follow a Gaussian distribution. If multiple sources are Gaussian, any linear combination of them results in another Gaussian distribution, rendering the original signals rotationally symmetric and impossible to uniquely separate. This fundamental limitation naturally translates the challenge of finding independent components into an optimization problem: we must find the projections that maximize non-Gaussianity.

The theoretical justification for this approach is deeply rooted in the Central Limit Theorem (CLT). The CLT establishes that the sum or any linear combination of independent random variables tends toward a Gaussian distribution, regardless of the distributions of the original variables. Consequently, any observed mixture

of independent, non-Gaussian sources will inherently appear more Gaussian than the constituent sources themselves. Therefore, by extracting the components that are the least Gaussian, we are effectively isolating the original independent signals.

1.3 Information Theory, Thermodynamics and CLT

The use of non-gaussianity as a measure for independence of components is also motivated by information theory. For a continuous random variable with a fixed variance, it is a foundational result that the Gaussian distribution uniquely possesses the maximum possible Shannon entropy. This mathematical property reveals a profound intrinsic link between information theory, the Central Limit Theorem, and statistical mechanics.

In thermodynamics, isolated physical systems naturally evolve toward states of maximum entropy, or maximum disorder. Viewed through this interdisciplinary lens, the Central Limit Theorem serves as the statistical manifestation of this physical principle. The mathematical process of linearly mixing independent sources inherently increases the overall entropy of the system, inevitably driving the joint distribution toward the high-entropy Gaussian state.

Recovering the underlying independent components, then, is conceptually equivalent to reversing this thermodynamic process. It requires projecting the empirical data away from the "disordered" high-entropy Gaussian manifold, intentionally guiding it back toward the highly structured, lower-entropy state of the original independent signals.

1.4 Problem Statement and Research Objectives

In practice, the ICA pipeline begins by centering and whitening the observed mixture. Whitening is a crucial preprocessing step that removes linear correlations and scales the variances to unity. Once the data is whitened, the search for independent components is geometrically restricted to finding an optimal orthogonal rotation. Specifically, the problem reduces to searching the unit sphere of the Stiefel manifold, space of orthogonal matrices, for the rotation matrix that maximizes the non-Gaussianity of the projected data.

The primary novelty of this research lies in proposing and investigating Optimal Transport metric, the Wasserstein distance, as the objective function for this rotational search. Our objective is to find the orthogonal candidates on the unit sphere that maximize the Wasserstein distance to a standard normal distribution. While traditional ICA algorithms often rely on sample approximations of population

moments (such as kurtosis) or surrogate information-theoretic measures (such as negentropy approximations), the Wasserstein metric offers a distinct advantage: it evaluates the true global geometry of the empirical distribution using order statistics (sorted data), providing a highly robust, non-parametric measure of distance to Gaussianity.

1.5 Structure of the Thesis

The remainder of this thesis is structured as follows. Chapter 2 establishes the theoretical foundations of linear ICA, information geometry, and Optimal Transport. Chapter 3 formalizes the squared Wasserstein-ICA (W2-ICA) framework, proving that the squared Wasserstein (W2) distance rigorously bounds mixture non-Gaussianity. Chapter 4 details the algorithmic enhancements required to optimize this metric on the Stiefel manifold efficiently. Chapter 5 presents our experimental evaluation, demonstrating the failure modes of standard proxy optimization (such as FastICA) and proposing a robust ground cost modification to handle outliers. Finally, Chapter 6 concludes the thesis with a summary of findings and directions for future research.

Chapter 2

Theoretical Foundations

2.1 Linear Independent Component Analysis (ICA)

2.1.1 Notation and the Linear Mixing Model

The foundational assumption of the linear Independent Component Analysis model is that the observed signals are generated by a linear combination of statistically independent latent sources. Let the observed data be denoted by a random vector $\mathbf{X} \in \mathbb{R}^d$, and the latent independent sources by $\mathbf{S} \in \mathbb{R}^d$. The linear mixing model is formally defined as

$$\mathbf{X} = \mathbf{A}\mathbf{S} \quad (2.1)$$

where $\mathbf{A} \in \mathbb{R}^{d \times d}$ is an unknown, invertible mixing matrix (Hyvärinen et al., 2001). Expanded into vector notation, the model represents the observed variables as

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1d} \\ a_{21} & a_{22} & \cdots & a_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ a_{d1} & a_{d2} & \cdots & a_{dd} \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_d \end{bmatrix}. \quad (2.2)$$

The core objective of ICA is to estimate an unmixing matrix $\mathbf{B} \approx \mathbf{A}^{-1}$ such that the recovered components $\mathbf{Z} = \mathbf{B}\mathbf{X}$ are as statistically independent as possible, thereby reconstructing the original sources $\mathbf{S}$.

2.1.2 Identifiability and Ambiguities

For the linear ICA model to be identifiable, it is a strict mathematical requirement that at most one of the underlying sources s_i follows a Gaussian distribution. If two or more sources are Gaussian, any orthogonal transformation of those sources yields another set of independent Gaussian variables. Consequently, the mixing

matrix cannot be uniquely determined from the observed data. Furthermore, even when the non Gaussianity assumption is met, the ICA model contains two inherent ambiguities. First, the variances (energies) of the independent components cannot be determined, which leads to a scaling ambiguity. Second, the order of the sources is arbitrary, leading to a permutation ambiguity. These ambiguities are generally acceptable in practice because the structural shape and independence of the source distributions are preserved.

2.1.3 Centering and Whitening

To simplify the optimization problem, the observed data $\mathbf{X}$ is subjected to a two step preprocessing phase consisting of centering and whitening. Centering ensures that the data has a zero mean by subtracting the expectation

$$\mathbb{E}[\mathbf{X}] = \mathbf{0}. \quad (2.3)$$

Given centered data, the covariance matrix simplifies to $\text{Cov}(\mathbf{X}) = \mathbb{E}[\mathbf{X}\mathbf{X}^\top]$. Whitening is then applied to transform the observed vector into a new vector $\tilde{\mathbf{X}}$ whose components are uncorrelated and possess unit variance. This is achieved using the eigenvalue decomposition of the covariance matrix

$$\text{Cov}(\mathbf{X}) = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top \quad (2.4)$$

where $\mathbf{V}$ is the orthogonal matrix of eigenvectors and $\mathbf{\Lambda}$ is the diagonal matrix of eigenvalues λ_j . The whitening transformation matrix $\mathbf{W}$ is constructed by scaling the eigenvectors by the inverse square root of the eigenvalues

$$\mathbf{W} = \mathbf{\Lambda}^{-1/2}\mathbf{V}^\top. \quad (2.5)$$

Applying this transformation yields the whitened data $\tilde{\mathbf{X}} = \mathbf{W}\mathbf{X}$. The covariance of this new variable is the identity matrix

$$\text{Cov}(\tilde{\mathbf{X}}) = \mathbf{W}\mathbb{E}[\mathbf{X}\mathbf{X}^\top]\mathbf{W}^\top = \mathbf{\Lambda}^{-1/2}\mathbf{V}^\top\mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top\mathbf{V}\mathbf{\Lambda}^{-1/2} = \mathbf{I}_d. \quad (2.6)$$

After whitening, the search for the unmixing matrix is drastically simplified. If we define the unmixing operation on the whitened data as $\tilde{\mathbf{Z}} = \mathbf{U}\tilde{\mathbf{X}}$, the covariance of the recovered components is

$$\text{Cov}(\tilde{\mathbf{Z}}) = \mathbf{U}\text{Cov}(\tilde{\mathbf{X}})\mathbf{U}^\top = \mathbf{U}\mathbf{I}_d\mathbf{U}^\top = \mathbf{U}\mathbf{U}^\top. \quad (2.7)$$

Because the independent components must have unit variance, we require $\mathbf{U}\mathbf{U}^\top = \mathbf{I}_d$. This reveals that the unmixing matrix $\mathbf{U}$ must be perfectly orthogonal. The ICA problem is thus reduced to finding an optimal rotation on the Stiefel manifold.

2.1.4 The Insufficiency of Principal Component Analysis

Principal Component Analysis is frequently used to decorrelate data, which corresponds to finding the matrix $\mathbf{V}^\top$ in the eigenvalue decomposition. However, making a sample decorrelated and whitened is not sufficient for source separation when the data is non Gaussian. Once the data is whitened such that $\mathbb{E}[\mathbf{X}\mathbf{X}^\top] = \mathbf{I}_d$, any rotational orientation satisfies the PCA criteria. For multivariate Gaussian distributions, decorrelation directly implies independence, leaving nothing further for PCA to accomplish. For non Gaussian data, however, higher order dependencies remain. ICA must therefore traverse the space of orthogonal matrices to find local optima of contrast functions such as $f(\mathbf{u}) = \mathbb{E}[(\mathbf{u}^\top \mathbf{X})^4]$ on the unit sphere S_n . Gradient descent and other optimization techniques are required to pinpoint the exact rotation that maximizes non Gaussianity and achieves full statistical independence.

2.2 An Information Geometry Perspective on ICA

2.2.1 Product and Gaussian Manifolds

Information geometry provides a rigorous framework for quantifying the relationship between independence and non Gaussianity by analyzing probability distributions as points on geometric manifolds. Following the perspective introduced by Cardoso (2022), ICA can be understood as projecting the empirical distribution onto two distinct subspaces. The first is the Product manifold $\mathcal{P}$, which contains all distributions where the components are strictly independent. The second is the Gaussian manifold $\mathcal{G}$, which encompasses all multivariate Gaussian distributions with zero mean and a fixed covariance structure. Standard decorrelation merely ensures that the covariance is diagonal, but full ICA operates at the intersection of these manifolds to find a linear transform that projects the empirical data as closely as possible onto the Product manifold (Cardoso, 2022).

2.2.2 Kullback-Leibler Pythagorean Identities

Distances between these distributions are measured using the Kullback-Leibler (KL) divergence. For any random vector Y with a joint density P_Y , the mutual information $\mathcal{I}(Y)$ measures the distance to the closest independent product

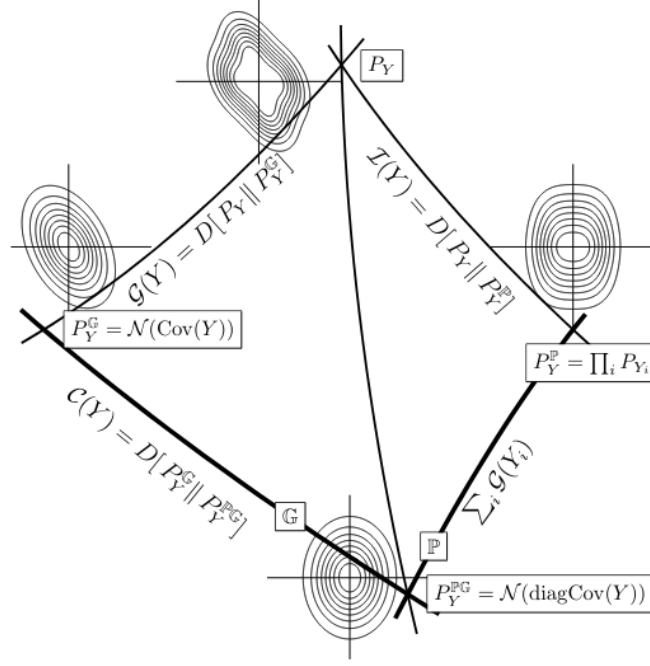


Figure 2.1: The Pythagorean geometry of Independent Component Analysis. The empirical joint distribution P_Y is orthogonally projected onto the Product manifold $\mathcal{P}$ (yielding the independent marginals P_Y^P) and the Gaussian manifold $\mathcal{G}$ (yielding the best Gaussian fit P_Y^G). The intersection of these approximations yields the Product-Gaussian distribution P_Y^{PG} . The Kullback-Leibler divergences between these nodes geometrically represent distinct statistical properties: Mutual Information $\mathcal{I}(Y) = D_{\text{KL}}(P_Y \| P_Y^P)$, Joint Non-Gaussianity $\mathcal{G}(Y) = D_{\text{KL}}(P_Y \| P_Y^G)$, and Correlation $\mathcal{C}(Y) = D_{\text{KL}}(P_Y^G \| P_Y^{PG})$. Figure adapted from (Cardoso, 2022).

distribution

$$\mathcal{I}(Y) = D_{\text{KL}} \left(P_Y \| \prod_i P_{Y_i} \right). \quad (2.8)$$

A Pythagorean identity exists on the Product manifold. For any candidate product distribution $Q = \prod_i q_i$, the divergence splits precisely into the mutual information and the marginal divergences

$$D_{\text{KL}}(P_Y \| Q) = \mathcal{I}(Y) + \sum_i D_{\text{KL}}(P_{Y_i} \| q_i). \quad (2.9)$$

A similar Pythagorean identity holds for the Gaussian manifold $\mathcal{G}$. The KL divergence from P_Y to any arbitrary Gaussian distribution $\mathcal{N}(\Sigma)$ decomposes into the divergence to the best Gaussian fit and the divergence between the two Gaussian models

$$D_{\text{KL}}(P_Y \| \mathcal{N}(\Sigma)) = D_{\text{KL}}(P_Y \| \mathcal{N}(\text{Cov } Y)) + D_{\text{KL}}(\mathcal{N}(\text{Cov } Y) \| \mathcal{N}(\Sigma)). \quad (2.10)$$

2.2.3 Relating Independence, Correlation, and Non-Gaussianity

These geometric identities can be combined to relate independence directly to non Gaussianity. We define the non Gaussianity of the joint variable Y as $G(Y) = D_{\text{KL}}(P_Y \parallel \mathcal{N}(\text{Cov } Y))$, and the scalar measure of correlation as $C(Y) = D_{\text{KL}}(\mathcal{N}(\text{Cov } Y) \parallel \mathcal{N}(\text{diag Cov } Y))$. Cardoso’s unified theorem states that

$$\mathcal{J}(Y) + \sum_i G(Y_i) = C(Y) + G(Y). \quad (2.11)$$

Crucially, the joint non Gaussianity $G(Y)$ is invariant under invertible linear transformations. Furthermore, after the data has been whitened, the covariance matrix is the identity matrix, meaning $\text{Cov}(Y) = \text{diag Cov}(Y)$ and therefore the correlation term $C(Y) = 0$. The relationship then drastically simplifies to

$$\mathcal{J}(Y) = G(Y) - \sum_i G(Y_i). \quad (2.12)$$

Because $G(Y)$ is constant under rotation, minimizing the mutual information $\mathcal{J}(Y)$ is mathematically equivalent to maximizing the sum of the marginal non Gaussianities $\sum_i G(Y_i)$. This rigorously proves that centering and whitening constrain the data to a subspace where true statistical independence can be achieved solely by maximizing non Gaussianity (Cardoso, 2022).

2.3 Optimal Transport Theory

2.3.1 The Monge-Kantorovich Problem

While information geometry utilizes Kullback-Leibler divergence, Optimal Transport offers an alternative metric space that is highly sensitive to the underlying geometry of the data. First formulated by Gaspard Monge (1781), the problem was originally conceptualized as finding the most cost effective way to move a distribution of mass, such as a pile of soil, to a target destination.

Mathematically, this is framed as finding a transport map T that pushes forward a source probability measure μ to a target measure ν , denoted as $T_{\#}\mu = \nu$. This means that for any measurable set A , the mass assigned by ν is equal to the mass assigned by μ to the preimage of A , such that $\nu(A) = \mu(T^{-1}(A))$. Because finding a deterministic map T is highly restrictive, Kantorovich (1942) later relaxed this formulation by introducing transport plans. This allowed mass from a single source location to be split and distributed across multiple targets. This relaxation transformed the rigorous mapping requirement into a joint probability distribution

problem, making the computation of transport costs mathematically tractable.

2.3.2 The Earth Mover’s Distance (W_1)

The minimal cost required to transform one probability distribution into another is formalized as the Wasserstein distance (Villani, 2003). The W_1 distance, commonly referred to as the Earth Mover’s Distance, quantifies the absolute effort of this transformation. For one dimensional distributions with cumulative distribution functions F and G , the W_1 distance is computed as the integral of the absolute differences between the cumulative functions

$$W_1(\mu, \nu) = \int_{-\infty}^{\infty} |F(x) - G(x)| dx. \quad (2.13)$$

Intuitively, W_1 sums the vertical area between the two distribution curves. While this metric is robust to extreme outliers, it often underestimates structural deviations in variance.

2.3.3 The Squared Wasserstein Distance (W_2)

To better capture variations in spread and central location, the Squared Wasserstein distance is utilized. It is most conveniently defined using the quantile functions, which are the inverses of the cumulative distribution functions F^{-1} and G^{-1} . The W_2 distance is formulated as

$$W_2(\mu, \nu) = \left(\int_0^1 |F^{-1}(t) - G^{-1}(t)|^2 dt \right)^{1/2}. \quad (2.14)$$

By squaring the transport cost, the W_2 metric heavily penalizes local geometric distortions. When evaluating the distance between an empirical projection and a standard normal distribution $\mathcal{N}(0, 1)$, W_2 provides a highly sensitive and reliable measure of Gaussianity. This makes it an ideal contrast function for Independent Component Analysis, as it maps the exact geometric deformations required to transform the candidate components into Gaussian noise.

2.3.4 Brenier’s Theorem and Caffarelli’s Regularity

The use of the squared Euclidean cost function in the W_2 distance provides profound geometric properties that are not present in W_1 . Specifically, Brenier’s Theorem establishes a direct link between the squared Wasserstein distance and convex analysis (Brenier, 1991). The theorem states that if the source measure μ is absolutely continuous with respect to the Lebesgue measure, the optimal transport

plan is uniquely given by a deterministic Monge map T . Crucially, Brenier proved that this optimal map is the gradient of a convex function ϕ , such that $T = \nabla\phi$.

The analytical behavior of this transport map is governed by Caffarelli's regularity theory (Caffarelli, 1992). Caffarelli demonstrated that if both the source and target probability densities are smooth and bounded strictly away from zero on their supports, the convex potential ϕ is strictly convex and its gradient T is continuously differentiable. This regularity implies that the optimal transport map does not tear or fold the probability space abruptly. In the context of linear ICA, the existence of a unique, smooth, and differentiable transport map mapping empirical projections to a Gaussian target provides the necessary mathematical foundation to analyze optimization gradients and theoretically bound mixture non-Gaussianities.

Chapter 3

The Squared Wasserstein-ICA (W2-ICA) Framework

3.1 Formulating ICA as an Optimal Transport Problem

As established in the preceding chapter, the process of centering and whitening restricts the search for independent components to the manifold of orthogonal matrices. Furthermore, Cardoso’s unified theorem (Cardoso, 2022) demonstrates that within this whitened space, the mutual information of the projections is minimized precisely when the sum of their marginal non-Gaussianities is maximized. Traditional Independent Component Analysis algorithms approach this optimization by utilizing proxy contrast functions, such as kurtosis or negentropy approximations. While computationally inexpensive, these proxies often capture only specific statistical moments, ignoring the broader geometric structure of the underlying probability distributions.

This thesis proposes a paradigm shift by formulating ICA directly as an Optimal Transport problem. Instead of relying on proxy moments, we utilize the squared Wasserstein distance, W_2^2 , as a direct, geometrically rigorous measure of non-Gaussianity. Let $\mathbf{X}$ be the whitened observed mixture, and let $\mathbf{u} \in S_n$ be a candidate projection vector on the unit sphere. The empirical distribution of the projected data is denoted by $\mathbb{P}_{\mathbf{u}^\top \mathbf{X}}$. We formally define the objective of the W2-ICA framework as finding the orthogonal rotation matrix whose column vectors $\mathbf{u}$ maximize the transport cost to a standard normal distribution $\mathcal{N}(0, 1)$:

$$\hat{\mathbf{u}} = \arg \max_{\mathbf{u} \in S_n} W_2^2(\mathbb{P}_{\mathbf{u}^\top \mathbf{X}}, \mathcal{N}(0, 1)). \quad (3.1)$$

By framing the problem in this manner, we leverage the Optimal Transport metric’s

unique ability to capture global geometric deviations from Gaussianity, providing a highly sensitive contrast function.

3.2 Theoretical Motivation: Bounding Mixture Non-Gaussianity

To mathematically justify the use of W_2^2 as an objective function for ICA, we must prove that maximizing this distance recovers the true independent sources. By the Central Limit Theorem, we possess the intuitive understanding that a mixture of non-Gaussian sources is more Gaussian than the constituent sources themselves. In the language of Optimal Transport, this implies that the Wasserstein distance between a mixture and a Gaussian distribution must be strictly bounded by the distances of its independent components.

3.2.1 Constructing the Common Source Coupling

To formally prove this bound, we construct a specific joint probability distribution, or coupling, using a common source of randomness. Let $\mathbf{N} = (N_1, \dots, N_d)^\top$ be a vector of d independent standard normal variables, such that $\mathbf{N} \sim \mathcal{N}(0, \mathbf{I}_d)$. For each true independent latent source Z_i , there exists an Optimal Transport map T_i that pushes forward the standard normal distribution to the source distribution, denoted as $(T_i)_\# \mathcal{N}(0, 1) = Z_i$.

We can construct a random variable X representing a linear mixture of the sources using a weight vector $\mathbf{w}$ defined on the unit sphere ($\sum_{i=1}^d w_i^2 = 1$). By applying the transport maps to our common noise vector, the mixture is defined as

$$X = \sum_{i=1}^d w_i T_i(N_i). \quad (3.2)$$

Because $T_i(N_i)$ follows the distribution of Z_i , it follows that X perfectly models the weighted mixture $\mathbf{w} \cdot \mathbf{Z}$. To compare this mixture to a Gaussian distribution, we construct a target variable Y using the exact same underlying noise vector $\mathbf{N}$ and the same weights:

$$Y = \sum_{i=1}^d w_i N_i. \quad (3.3)$$

Given that the weights sum to unity and the components N_i are independent standard normals, the resulting variable Y is guaranteed to follow a standard normal distribution, $Y \sim \mathcal{N}(0, 1)$. The pair (X, Y) therefore creates a valid coupling $\pi_{X,Y}^{\mathbf{N}}$ between the mixture distribution and the Gaussian target.

3.2.2 Proof of the W_2 Upper Bound

The squared Wasserstein distance is defined as the infimum of the expected squared cost over all possible valid couplings. Because the infimum represents the absolute optimal transport plan, the cost of the optimal plan must be less than or equal to the cost of our specifically constructed coupling (X, Y) . This yields the fundamental inequality:

$$W_2^2(\mathbf{w} \cdot \mathbf{Z}, \mathcal{N}(0, 1)) \leq \mathbb{E} [|X - Y|^2]. \quad (3.4)$$

We evaluate the right side of this inequality by expanding the squared difference:

$$\begin{aligned} |X - Y|^2 &= \left| \sum_{i=1}^d w_i T_i(N_i) - \sum_{i=1}^d w_i N_i \right|^2 \\ &= \left(\sum_{i=1}^d w_i (T_i(N_i) - N_i) \right)^2. \end{aligned} \quad (3.5)$$

Squaring the summation produces a combination of diagonal terms where $i = j$ and cross terms where $i \neq j$:

$$|X - Y|^2 = \sum_{i=1}^d w_i^2 (T_i(N_i) - N_i)^2 + \sum_{i \neq j} w_i w_j (T_i(N_i) - N_i) (T_j(N_j) - N_j). \quad (3.6)$$

Taking the mathematical expectation of this expansion drastically simplifies the expression. Because the underlying noise variables N_i and N_j are independent, and the functions evaluate to a mean of zero, the expectation of all cross terms strictly vanishes. We are left solely with the expectation of the diagonal terms:

$$\mathbb{E} [|X - Y|^2] = \sum_{i=1}^d w_i^2 \mathbb{E} [(T_i(N_i) - N_i)^2]. \quad (3.7)$$

Recognizing that $\mathbb{E}[(T_i(N_i) - N_i)^2]$ is the exact definition of the squared Wasserstein distance between the individual source Z_i and a standard normal distribution, we arrive at the final upper bound:

$$W_2^2(\mathbf{w} \cdot \mathbf{Z}, \mathcal{N}(0, 1)) \leq \sum_{i=1}^d w_i^2 W_2^2(Z_i, \mathcal{N}(0, 1)). \quad (3.8)$$

This convexity adjacent property proves that mixing independent sources mathematically reduces the Wasserstein distance to Gaussianity. Maximizing the left-hand side forces the weight vector $\mathbf{w}$ to collapse toward a canonical axis, effectively isolating a single independent source.

3.2.3 Comonotonicity and the Proof of Strict Inequality

While the upper bound demonstrates that mixtures do not exceed the non-Gaussianity of their sources, successfully optimizing ICA requires proving that mixtures are strictly less non-Gaussian than the original sources. We must therefore determine the conditions under which the aforementioned inequality becomes a strict inequality.

In a one-dimensional setting, the W_2 distance equals the expected squared difference of a specific coupling if and only if the random variables are comonotonic. By Brenier's Theorem (Brenier, 1991), this implies there exists a strictly increasing, deterministic function mapping one variable to the other. For our constructed mixture variable X and the Gaussian target Y to be comonotonic with respect to the shared noise vector $\mathbf{N} \in \mathbb{R}^d$, their gradients must be perfectly parallel at every point in the probability space. This requires the existence of a scalar scaling function $\lambda(\mathbf{N})$ such that:

$$\nabla X(\mathbf{N}) = \lambda(\mathbf{N}) \nabla Y(\mathbf{N}). \quad (3.9)$$

Assuming the source distributions possess smooth and strictly positive densities, Caffarelli's regularity theory (Caffarelli, 1992) guarantees that the Optimal Transport maps T_i are continuously differentiable. We can therefore compute the gradients of our constructed variables with respect to the noise vector $\mathbf{N}$. The gradient of the Gaussian target is simply the weight vector:

$$Y(\mathbf{N}) = \sum_{i=1}^d w_i N_i \implies \nabla Y = \mathbf{w}. \quad (3.10)$$

The gradient of the source mixture leverages the derivatives of the individual transport maps:

$$X(\mathbf{N}) = \sum_{i=1}^d w_i T_i(N_i) \implies \nabla X = \begin{bmatrix} w_1 T'_1(N_1) \\ w_2 T'_2(N_2) \\ \vdots \\ w_d T'_d(N_d) \end{bmatrix}. \quad (3.11)$$

Substituting these gradients back into our comonotonicity condition yields the system of equations:

$$\nabla X = \lambda(\mathbf{N}) \mathbf{w}. \quad (3.12)$$

To determine if this condition can hold for non-Gaussian sources, we differentiate both the Left-Hand Side (LHS) and the Right-Hand Side (RHS) with respect to $\mathbf{N}$

to compute their respective Hessian (Jacobian) matrices.

On the left hand side, because X separates into independent terms $w_i T_i(N_i)$, the cross-derivatives $\frac{\partial^2 X}{\partial N_i \partial N_j}$ are strictly zero for all $i \neq j$. This yields a purely diagonal matrix containing the second derivatives of the transport maps.

On the right hand side, we differentiate the product $\lambda(\mathbf{N})\mathbf{w}$. Because the weight vector $\mathbf{w}$ is a constant, applying the product rule yields the outer product of $\mathbf{w}$ and the gradient of the scalar function $\nabla\lambda$. This forms a Rank-1 matrix.

Equating the left and right hand sides:

$$\underbrace{\begin{bmatrix} w_1 T_1''(N_1) & 0 & \cdots & 0 \\ 0 & w_2 T_2''(N_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & w_d T_d''(N_d) \end{bmatrix}}_{\text{Hessian of } X \text{ (Diagonal)}} = \underbrace{\begin{bmatrix} w_1 \frac{\partial \lambda}{\partial N_1} & w_1 \frac{\partial \lambda}{\partial N_2} & \cdots & w_1 \frac{\partial \lambda}{\partial N_d} \\ w_2 \frac{\partial \lambda}{\partial N_1} & w_2 \frac{\partial \lambda}{\partial N_2} & \cdots & w_2 \frac{\partial \lambda}{\partial N_d} \\ \vdots & \vdots & \ddots & \vdots \\ w_d \frac{\partial \lambda}{\partial N_1} & w_d \frac{\partial \lambda}{\partial N_2} & \cdots & w_d \frac{\partial \lambda}{\partial N_d} \end{bmatrix}}_{\text{Outer Product } \mathbf{w}(\nabla\lambda)^\top \text{ (Rank-1)}} \quad (3.13)$$

This matrix equality forces a profound contradiction. Let us examine any off-diagonal entry located at row i and column j (where $i \neq j$). The diagonal LHS matrix dictates that this entry must be exactly zero. However, the RHS Rank-1 matrix evaluates this same entry as $w_i \frac{\partial \lambda}{\partial N_j}$. This establishes the requirement:

$$w_i \frac{\partial \lambda}{\partial N_j} = 0 \quad \text{for all } i \neq j. \quad (3.14)$$

Because we are assuming a valid mixture where the component weights are non-zero ($w_i \neq 0$), it must be true that $\frac{\partial \lambda}{\partial N_j} = 0$. Since this holds across all dimensions, the gradient of the scaling function is zero ($\nabla\lambda = \mathbf{0}$). Consequently, $\lambda(\mathbf{N})$ must be a global scalar constant, λ .

If λ is a constant, then equating the diagonal elements of our matrices implies that $T_i'(N_i) = \lambda$. Integrating this relationship means the Optimal Transport maps themselves must be strictly linear functions of the form:

$$T_i(x) = \lambda x + c. \quad (3.15)$$

We know that applying a purely linear transformation to a Gaussian noise source N_i can only ever produce another Gaussian distribution. However, the foundational identifiability assumption of ICA guarantees that our original latent sources Z_i are non-Gaussian. Because the sources are non-Gaussian, their Optimal Transport mappings from a Gaussian base must be non-linear (curved).

Therefore, the linear map requirement derived from the comonotonicity condition fundamentally fails for non-Gaussian sources. Because the condition for

equality cannot be met, we conclude that for any valid mixture of independent, non-Gaussian variables, the relationship is a strict inequality:

$$W_2^2(\mathbf{w} \cdot \mathbf{Z}, \mathcal{N}(0, 1)) < \sum_{i=1}^d w_i^2 W_2^2(Z_i, \mathcal{N}(0, 1)). \quad (3.16)$$

3.3 Empirical Landscape: W_1 versus W_2

Having theoretically established that the squared Wasserstein distance successfully bounds mixture non-Gaussianity, we now evaluate its practical suitability as an optimization landscape. In the context of ICA, the objective is to *maximize* non-Gaussianity. Therefore, gradient ascent algorithms must locate the peaks (maxima) of the distance metric to successfully isolate the independent sources. We compare the empirical gradients of the squared W_2 distance (Mean Squared Error ground cost) against the W_1 distance (Mean Absolute Error ground cost) to observe their behavior during rotational search.

3.3.1 Student-t Distributions and Degrees of Freedom

To conduct this empirical comparison, we generated synthetic bivariate mixtures from Student-t distributions. The Student-t distribution is ideal for this analysis because its non-Gaussianity can be precisely controlled by adjusting its degrees of freedom (ν). A low degrees of freedom parameter (e.g., $\nu = 2$) produces heavy tails, representing a strong non-Gaussian signal. Conversely, as ν approaches higher values (e.g., $\nu = 10$), the tails become lighter, and the distribution closely approximates a standard normal, representing a weak non-Gaussian signal.

We systematically evaluated the W_1 and W_2^2 distances from a standard normal distribution across a continuous range of rotation angles $\theta \in [0, \pi]$. Because the whitened unmixing matrix restricts the search to orthogonal vectors, the two true independent components in a bivariate mixture are geometrically separated by exactly $\pi/2$ radians (90°). Consequently, sweeping the projection vector across a π interval should theoretically reveal exactly two distinct maxima, corresponding to the two latent sources.

3.3.2 Robustness to Weak Non-Gaussian Signals

The empirical results, visualized in Figure 3.1, reveal a stark contrast in the optimization landscapes of the two metrics. For strong non-Gaussian signals ($\nu = 2$), both metrics successfully identify the global optima, exhibiting two distinct peaks separated by $\pi/2$. However, the W_2^2 metric consistently exhibits

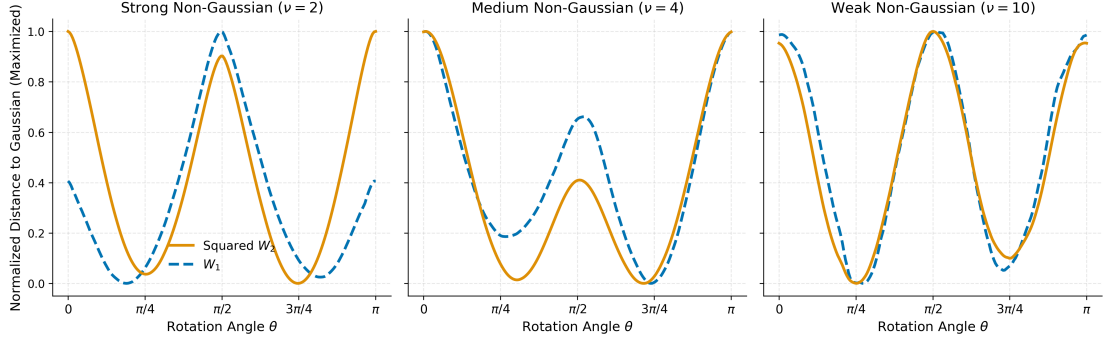


Figure 3.1: Comparison of the optimization landscapes for the W_1 and W_2^2 metrics across varying signal strengths. Distances are normalized to a $[0, 1]$ scale for direct shape comparison. The two peaks represent the maximum non-Gaussianity locations of the two independent components, separated by $\pi/2$ radians. While both metrics perform well for strong signals, the W_1 landscape deteriorates into flat plateaus for weak signals, whereas W_2^2 maintains smooth, strictly guiding gradients.

significantly higher curvature (strict concavity) around these optima, providing a stronger, more directional gradient for optimization solvers to climb.

The critical divergence in algorithmic stability occurs in the presence of weak signals ($\nu = 10$). As the underlying distribution approaches Gaussian noise, the structural deviations are confined almost entirely to the extreme tails. The W_1 metric, relying on absolute error, fails to sufficiently penalize these tail deviations. Consequently, the W_1 landscape degrades into a flat, jagged plateau around the true sources. This flatness causes gradients to vanish prematurely, virtually guaranteeing that optimization solvers will stall or become trapped in spurious local noise.

In contrast, the squared penalty inherent to the W_2^2 metric heavily magnifies these subtle tail deviations. The W_2^2 metric remains remarkably smooth and unimodal around the optimum, maintaining a strict concavity that guides the solver to the true peak despite the weak signal strength. This robust gradient retention justifies the computational expense of selecting the squared Wasserstein distance as the core objective function for the proposed W2-ICA framework.

Chapter 4

Algorithmic Enhancements

4.1 The Baseline W2-ICA Algorithm

Having established the theoretical optimality of the squared Wasserstein distance (W_2^2) for Independent Component Analysis, the challenge transitions to computationally minimizing this cost function. The baseline W2-ICA algorithm proceeds by first centering and whitening the observed mixture $\mathbf{X} \in \mathbb{R}^{d \times N}$, yielding the whitened data $\tilde{\mathbf{X}}$. The algorithm then initializes an orthogonal unmixing matrix $\mathbf{W} \in \mathbb{R}^{d \times d}$. In the forward pass, the data is projected onto the candidate components, $\mathbf{Y} = \mathbf{W}\tilde{\mathbf{X}}$. For each row (component) of $\mathbf{Y}$, the empirical cumulative distribution function is constructed by sorting the N observations. The W_2^2 distance is then computed by matching these sorted empirical quantiles to the corresponding quantiles of a standard normal distribution.

While theoretically sound, the naive implementation of this gradient descent loop faces severe computational and geometric hurdles. The exact Optimal Transport matching requires sorting at every iteration, presenting a significant computational bottleneck. Furthermore, standard gradient updates do not inherently respect the strict orthogonality constraint of the mixing matrix, and discrete approximations of the target Gaussian introduce gradient noise. To make W2-ICA computationally competitive with existing methods, several specific algorithmic and geometric enhancements are required.

4.2 Computational Bottlenecks and Solutions

4.2.1 Exact Analytical Gaussian Targets

A primary source of instability in the baseline algorithm is the discrete approximation of the target Gaussian distribution. Traditionally, the target quantiles are approximated by evaluating the inverse cumulative distribution function at

evenly spaced intervals, such as $q_i = \Phi^{-1}((i - 0.5)/N)$. However, representing a continuous, infinite-tailed Gaussian distribution with a finite set of discrete points introduces approximation noise. When the algorithm computes the gradient of the W_2^2 distance, this noise translates into a jagged optimization landscape, severely hindering convergence as the components approach true independence.

To eliminate this approximation noise, we replace the point-sampled targets with an exact analytical formulation. Using integral calculus, we compute the exact expected value of a standard normal variable within each discrete quantile bin. Let the uniform probability bin edges be defined as $p_i = \frac{i}{N}$ for $i = 0, \dots, N$, with corresponding Gaussian domain boundaries $z_i = \Phi^{-1}(p_i)$. The exact analytical target value T_i for the i -th sorted sample is computed as the conditional expectation within that interval:

$$T_i = N \int_{z_{i-1}}^{z_i} x \phi(x) dx \quad (4.1)$$

where $\phi(x)$ is the standard normal probability density function. Evaluating this integral yields the closed-form solution:

$$T_i = N(\phi(z_{i-1}) - \phi(z_i)). \quad (4.2)$$

By computing this exact analytical target once during initialization, the algorithm evaluates the W_2^2 distance against a mathematically perfect discrete representation of a Gaussian, ensuring perfectly smooth gradients near the global optimum.

4.2.2 Batched Vectorization for Restarts

Unlike Principal Component Analysis, which yields a unique global solution via singular value decomposition, the ICA optimization landscape is highly non-convex and characterized by numerous local optima. Consequently, identifying the true independent components strictly requires multiple random initializations (restarts) to thoroughly explore the hypothesis space.

Because evaluating the W_2^2 objective inherently requires an $\mathcal{O}(N \log N)$ sorting operation for each candidate component, executing these random restarts sequentially via standard iterative loops is computationally prohibitive. To resolve this, we utilize a batched vectorization strategy. Rather than maintaining a single weight matrix, we aggregate B random restarts into a single high-dimensional tensor $\mathbf{W}_{\text{batch}} \in \mathbb{R}^{B \times d \times d}$.

This formulation allows the projection $\mathbf{Y} = \mathbf{W}_{\text{batch}} \tilde{\mathbf{X}}$ to be computed simultaneously for all B trajectories. The elements of the resulting batch tensor can be sorted in parallel. The pre-computed exact analytical target vector $\mathbf{T}$ is subsequently broadcast-subtracted across the entire batch to compute the distances and

gradients in a single step.

4.3 Optimization on the Stiefel Manifold

4.3.1 The Riemannian Gradient

Because the input data is whitened, the unmixing matrix $\mathbf{W}$ must remain strictly orthogonal, satisfying $\mathbf{W}\mathbf{W}^\top = \mathbf{I}$. The geometric space of all such orthogonal matrices forms the Stiefel manifold, denoted $\mathcal{S}$. If we compute the standard Euclidean gradient of the squared Wasserstein distance, $\mathbf{G} = \nabla_{\mathbf{W}} W_2^2$, this vector points into the flat, unconstrained ambient space $\mathbb{R}^{d \times d}$. Updating the matrix using this Euclidean gradient would immediately violate the orthogonality constraint.

To navigate the curved geometry of the Stiefel manifold, we must project the Euclidean gradient onto the tangent space of the manifold at the current point $\mathbf{W}$, as illustrated in Figure 4.1.

The resulting Riemannian gradient $\nabla_{\mathcal{S}} \mathbf{W}$ is given by (Absil, Mahony, & Sepulchre, 2008):

$$\nabla_{\mathcal{S}} \mathbf{W} = \mathbf{G} - \frac{1}{2}(\mathbf{G}\mathbf{W}^\top + \mathbf{W}\mathbf{G}^\top)\mathbf{W}. \quad (4.3)$$

Geometrically, the term $(\mathbf{G}\mathbf{W}^\top + \mathbf{W}\mathbf{G}^\top)$ precisely measures the symmetric violation of orthogonality induced by the Euclidean gradient. By subtracting this specific violation, we ensure that the update step perfectly hugs the curvature of the

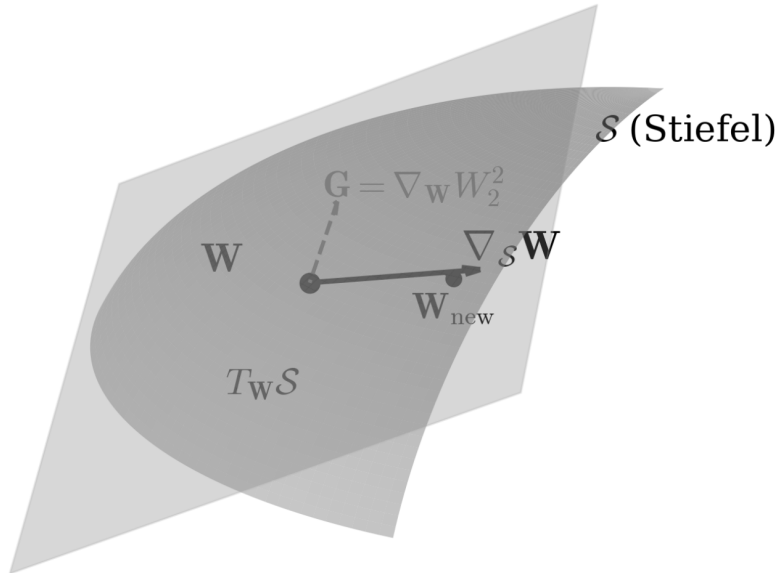


Figure 4.1: Geometric sketch of optimization on the Stiefel manifold $\mathcal{S}$. The Euclidean gradient $\mathbf{G}$ is projected onto the tangent space $T_{\mathbf{W}}\mathcal{S}$ to yield the Riemannian gradient $\nabla_{\mathcal{S}} \mathbf{W}$. Following a step along the tangent plane, a retraction maps the estimate back to the orthogonal manifold surface at $\mathbf{W}_{\text{new}}$.

manifold, preserving the decorrelation of the components.

4.3.2 Retraction via Symmetric Decorrelation

Although the Riemannian gradient ensures the update step is tangential to the Stiefel manifold, moving along this flat tangent plane for a finite step size η will cause the new matrix $\mathbf{W}_{\text{step}} = \mathbf{W} - \eta \nabla_{\mathcal{S}} \mathbf{W}$ to slightly lift off the curved manifold surface. To restore exact orthogonality, the matrix must be mapped back onto the manifold via a mathematical operation known as a retraction (Absil et al., 2008).

While the Gram-Schmidt orthogonalization process is a common approach, it is fundamentally asymmetric; it permanently fixes the first vector and modifies subsequent vectors to fit, which inherently biases simultaneous optimization. Instead, we utilize symmetric decorrelation as our retraction mechanism:

$$\mathbf{W}_{\text{new}} = (\mathbf{W}_{\text{step}} \mathbf{W}_{\text{step}}^{\top})^{-1/2} \mathbf{W}_{\text{step}}. \quad (4.4)$$

This operation computes the overlap covariance of the stepped matrix and applies a uniform, symmetric adjustment to all vectors simultaneously. This ensures that no single independent component is arbitrarily prioritized during the retraction process, maintaining the stability of the batched optimization.

4.4 Navigating Discrete Optimization Landscapes

While the W_2 -ICA framework demonstrates robust convergence on continuous distributions, real-world data often contains discrete signals, such as binary states, digital communications, or count-based market events. Applying continuous Optimal Transport directly to discrete mixtures introduces severe geometric pathologies that fracture the optimization landscape.

4.4.1 The Shattered Landscape of Discrete CDFs

The exact analytical Gaussian target developed in Section 4.2 relies on matching the empirical cumulative distribution function (CDF) of the projected data. For a continuous variable, this empirical CDF smoothly approximates a strictly increasing curve. However, for highly discrete data, any one-dimensional projection creates a rigid, step-like "staircase" CDF.

This discreteness breaks the continuous geometry of the W_2 metric due to the fundamental mechanics of sorting. Evaluating the Wasserstein distance requires sorting the projected points. As the optimization algorithm rotates the unmixing matrix $\mathbf{W}$, dense clumps of discrete data points slide past one another, causing

the sorting order to jump abruptly. Every instantaneous change in the sorting permutation causes a discontinuous shift in the gradient direction. Consequently, the theoretically smooth W_2 landscape shatters into a highly jagged, non-differentiable surface filled with combinatorial local optima. The optimizer becomes permanently trapped on spurious peaks (often corresponding to diagonal projections of the discrete hypercube), leading to catastrophic unmixing failures.

4.4.2 Continuous Smoothing via Gaussian Dithering

To restore the smooth gradients of the Stiefel manifold without resorting to computationally prohibitive $\mathcal{O}(N^2)$ entropic regularization techniques (such as Sinkhorn distances), we introduce an algorithmic Gaussian Dithering step.

Adapted from classical signal processing techniques used to decorrelate quantization error (Schuchman, 1964), we inject a microscopic amount of continuous, zero-mean Gaussian noise ($\sigma_{\text{dither}} \approx 0.01$) into the projected components immediately prior to the sorting operation in the forward pass. Mathematically, this operation convolves the discrete Dirac spikes of the empirical distribution with a narrow Gaussian kernel, as illustrated in Figure 4.2.

This deforms the rigid staircase of the discrete CDF into a strictly increasing, smooth continuous curve. By eliminating non-differentiable sorting ties, Gaussian dithering restores stable, informative gradients while preserving the high-speed $\mathcal{O}(N \log N)$ computational efficiency of the 1D Optimal Transport formulation.

It is crucial to emphasize that this dithering process does not constitute the injection of restrictive prior knowledge regarding the source distributions. Rather, it acts as a blind, universal mathematical regularizer applied uniformly to the 1D projections $\mathbf{w}^\top \mathbf{X}$. For latent sources that are already continuous, the addition of microscopic variance ($\sigma_{\text{dither}} \approx 0.01$) is statistically negligible and preserves the underlying Wasserstein geometry. For discrete sources, it prevents catastrophic sorting failures. Furthermore, perfectly discrete probability masses exist almost exclusively in synthetic simulations; real-world digital signals inherently possess continuous baseline sensor noise or measurement inaccuracies. The Gaussian dithering step safely mimics this natural continuous noise floor, ensuring the optimization manifold remains globally differentiable across all data modalities.

4.4.3 Escaping Traps with Stochastic Mini-Batching

Even with a smoothed CDF, the global optimization landscape of a discrete mixture remains highly non-convex. Evaluating the exact W_2 distance using the full dataset guarantees deterministic convergence to the *nearest* local maximum.

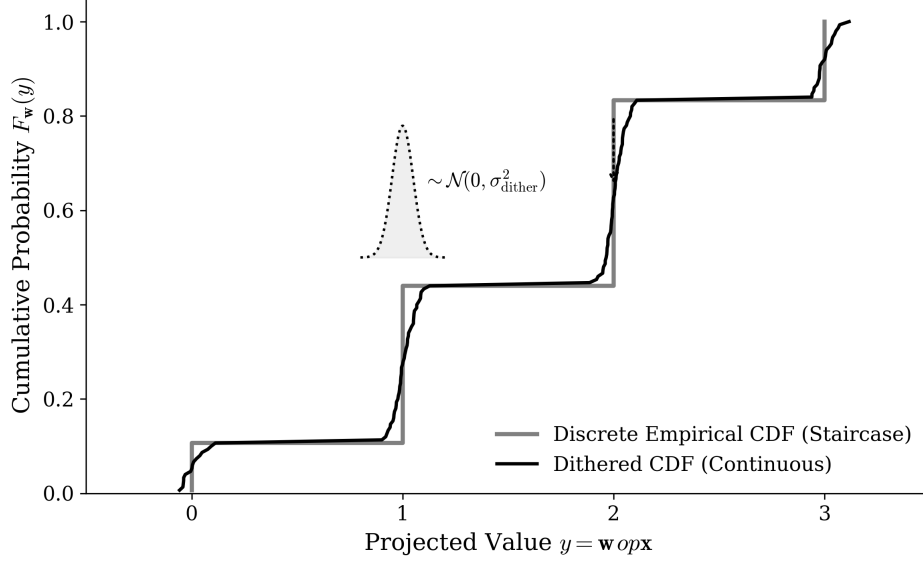


Figure 4.2: The effect of algorithmic Gaussian Dithering. The rigid, non-differentiable staircase of a discrete empirical CDF (gray) is mathematically convolved with a narrow Gaussian kernel, yielding a strictly increasing, smooth continuous curve (black) that restores stable Wasserstein gradients.

In an environment riddled with shallow discrete traps, this deterministic behavior is fatal.

To counter this, we transition from full-batch evaluation to Stochastic Mini-Batching. At each optimization step, the algorithm evaluates the Wasserstein gradients using a randomly sampled subset of the data (e.g., $N_{\text{batch}} = 512$ or 1024). This stochastic subsampling introduces calculated gradient noise, acting as a form of simulated annealing. A geometric trap that exists for the full dataset may not exist, or may have a significantly different gradient profile, for a specific random subset. This stochastic momentum allows the optimizer to "bounce" out of spurious local minima. The combination of Gaussian dithering to smooth the discrete steps and stochastic mini-batching to navigate the non-convex topology allows W2-ICA to successfully separate mixed environments.

4.5 W2-ICA Algorithm Overview

The individual theoretical components and algorithmic enhancements detailed above are synthesized into the final W2-ICA training procedure. Algorithm 1 outlines the complete pseudo-code for extracting independent non-Gaussian components from a mixed dataset using stochastic Riemannian optimization.

Algorithm 1 The Squared Wasserstein-ICA (W2-ICA) Algorithm

Require: Observed mixture $\mathbf{X} \in \mathbb{R}^{d \times N}$, iterations K , learning rate η , batch size B , dither noise σ_{dither}

Ensure: Estimated unmixing matrix $\mathbf{W}_{\text{final}} \in \mathbb{R}^{d \times d}$

1: **Phase 1: Preprocessing & Target Generation**

2: Center the data: $\mathbf{X} \leftarrow \mathbf{X} - \mathbb{E}[\mathbf{X}]$

3: Whiten the data: $\tilde{\mathbf{X}} \leftarrow (\mathbf{X}\mathbf{X}^\top)^{-1/2}\mathbf{X} \triangleright$ Yields unit variance, uncorrelated data

4: Compute analytical target $\mathbf{T} \in \mathbb{R}^B$ via exact Gaussian CDF integration

5: **Phase 2: Initialization**

6: Initialize $\mathbf{W} \in \mathbb{R}^{d \times d}$ randomly from $\mathcal{N}(0, 1)$

7: Apply symmetric decorrelation:

$$\mathbf{W} \leftarrow (\mathbf{W}\mathbf{W}^\top)^{-1/2}\mathbf{W} \quad \triangleright \text{Initialize on Stiefel manifold}$$

8: **Phase 3: Stochastic Riemannian Optimization**

9: **for** $k = 1$ to K **do**

10: Sample stochastic mini-batch $\tilde{\mathbf{X}}_{\text{batch}} \in \mathbb{R}^{d \times B}$ from $\tilde{\mathbf{X}}$

11: Project data onto candidates: $\mathbf{Y} \leftarrow \mathbf{W}\tilde{\mathbf{X}}_{\text{batch}}$

12: Apply Gaussian Dithering: $\mathbf{Y}_{\text{dither}} \leftarrow \mathbf{Y} + \mathcal{N}(0, \sigma_{\text{dither}}^2)$

13: Sort each row of $\mathbf{Y}_{\text{dither}}$ ascending to yield empirical quantiles $\mathbf{Y}_{\text{sorted}}$

14: Compute Wasserstein objective (maximize non-Gaussianity):

$$\mathcal{L} = \frac{1}{B} \sum_{i=1}^d \|\mathbf{Y}_{\text{sorted}}^{(i)} - \mathbf{T}\|_2^2$$

15: Backpropagate to compute Euclidean gradient: $\mathbf{G} = \nabla_{\mathbf{W}} \mathcal{L}$

16: Project to tangent space (Riemannian Gradient):

$$\nabla_{\mathcal{S}} \mathbf{W} = \mathbf{G} - \frac{1}{2}(\mathbf{G}\mathbf{W}^\top + \mathbf{W}\mathbf{G}^\top)\mathbf{W}$$

17: Update matrix along tangent plane:

$$\mathbf{W}_{\text{step}} = \mathbf{W} + \eta \nabla_{\mathcal{S}} \mathbf{W} \quad \triangleright \text{Gradient Ascent}$$

18: Retract to Stiefel manifold via symmetric decorrelation:

$$\mathbf{W} \leftarrow (\mathbf{W}_{\text{step}}\mathbf{W}_{\text{step}}^\top)^{-1/2}\mathbf{W}_{\text{step}}$$

19: Optional: Apply learning rate decay $\eta \leftarrow \eta \cdot 0.99$

20: **end for**

21: **Phase 4: Finalization**

22: $\mathbf{W}_{\text{final}} \leftarrow \mathbf{W}(\mathbf{X}\mathbf{X}^\top)^{-1/2} \quad \triangleright \text{Map back to original data space}$

23: **return** $\mathbf{W}_{\text{final}}$

4.6 The Intractability of Fixed Point Algorithms for W2-ICA

The industry standard algorithm for ICA, FastICA (Hyvärinen et al., 2001), achieves highly efficient, super-linear convergence by employing a Newton (second-

order) fixed-point step. FastICA is mathematically able to utilize this approach because it maximizes proxy contrast functions (such as the *logcosh* function) that are static and easily differentiable. A natural question arises: can we formulate a "Fast-Wasserstein-ICA" by deriving an exact Newton step for the W_2^2 objective?

To implement a Newton step, we would require the exact Hessian matrix of the squared Wasserstein distance, $\mathbf{H} = \nabla_{\mathbf{w}}^2 W_2^2$. The first derivative (the gradient) of the objective with respect to the weight vector $\mathbf{w}$ utilizes the Optimal Transport map $T_{\mathbf{w}}$:

$$\nabla_{\mathbf{w}} \left(\frac{1}{2} W_2^2 \right) = \mathbb{E} \left[\mathbf{X} (\mathbf{w}^\top \mathbf{X} - T_{\mathbf{w}}(\mathbf{w}^\top \mathbf{X})) \right]. \quad (4.5)$$

To compute the Hessian, we must differentiate the optimal transport map $T_{\mathbf{w}}(\mathbf{w}^\top \mathbf{X})$ itself with respect to $\mathbf{w}$. Applying the total derivative yields two distinct terms:

$$\nabla_{\mathbf{w}} [T_{\mathbf{w}}(\mathbf{w}^\top \mathbf{X})] = \underbrace{T'_{\mathbf{w}}(\mathbf{w}^\top \mathbf{X}) \mathbf{X}}_{\text{Argument Derivative}} + \underbrace{(\nabla_{\mathbf{w}} T_{\mathbf{w}})(\mathbf{w}^\top \mathbf{X})}_{\text{Map Derivative}}. \quad (4.6)$$

By Caffarelli's regularity theory (Caffarelli, 1992), the transport map $T_{\mathbf{w}}$ is continuously differentiable, guaranteeing the existence of the Argument Derivative. The structural problem lies exclusively within the Map Derivative.

In the 1D Optimal Transport setting, the transport map to a standard Gaussian relies directly on the empirical cumulative distribution function, $F_{\mathbf{w}}$, of the projected data $y = \mathbf{w}^\top \mathbf{X}$. Expanding the Map Derivative yields:

$$(\nabla_{\mathbf{w}} T_{\mathbf{w}})(y) = \nabla_{\mathbf{w}} [\Phi^{-1}(F_{\mathbf{w}}(y))] = \frac{1}{\phi(\Phi^{-1}(F_{\mathbf{w}}(y)))} \nabla_{\mathbf{w}} F_{\mathbf{w}}(y). \quad (4.7)$$

This introduces a severe mathematical pitfall. The term $\nabla_{\mathbf{w}} F_{\mathbf{w}}(y)$ represents the rate of change of the cumulative probability mass as the projection plane $\mathbf{w}$ is rotated. Conceptually and mathematically, the derivative of a cumulative distribution function's boundary relies strictly on the underlying Probability Density Function (PDF) evaluated exactly at that boundary.

This requirement fundamentally defeats the primary advantage of the W2-ICA formulation. This advantage of 1D Optimal Transport stems from its reliance on sorting (empirical CDFs), completely circumventing the need for continuous density estimation. Requiring the PDF to compute the Hessian forces the algorithm to rely on density approximation methods like Kernel Density Estimation (KDE) or Normalizing Flows. These methods introduce statistical noise, hyperparameter tuning (e.g. bandwidth), and computational overhead.

Because the exact Hessian requires continuous density estimation, computing a true Newton-based fixed-point step for exact W2-ICA is mathematically intractable. Consequently, the optimal strategy for the W2-ICA framework is the utilization

of Quasi-Newton methods, specifically the Limited-memory BFGS (L-BFGS) algorithm (Liu & Nocedal, 1989) adapted for the Stiefel manifold (Absil et al., 2008). L-BFGS intelligently approximates the inverse Hessian curvature strictly by observing the history of the highly stable, first-order Wasserstein gradients. This approach bridges the gap, allowing the algorithm to achieve faster convergence by preserving the sorting-based methodology of the Optimal Transport formulation.

Chapter 5

Experimental Evaluation and Applications

5.1 Evaluation Metrics: The Amari Performance Index

To empirically evaluate the success of an Independent Component Analysis (ICA) algorithm, we require a quantitative metric that measures the distance between the estimated unmixing matrix $\mathbf{W}_{\text{est}}$ and the true mixing matrix $\mathbf{A}$. Because ICA is subject to inherent scaling and permutation ambiguities, we utilize the Amari Performance Index, often referred to as the Amari error (Amari, Cichocki, & Yang, 1996). The Amari error measures how close the global transfer matrix $\mathbf{P} = \mathbf{W}_{\text{est}}\mathbf{A}$ is to a generalized permutation matrix, a matrix containing exactly one non-zero element in each row and column. For a $d \times d$ matrix $\mathbf{P}$ with elements p_{ij} , the Amari error is defined as:

$$E(\mathbf{P}) = \frac{1}{2d} \sum_{i=1}^d \left(\sum_{j=1}^d \frac{|p_{ij}|}{\max_k |p_{ik}|} - 1 \right) + \frac{1}{2d} \sum_{j=1}^d \left(\sum_{i=1}^d \frac{|p_{ij}|}{\max_k |p_{kj}|} - 1 \right). \quad (5.1)$$

This formulation normalizes each row and column by its maximum absolute value. If $\mathbf{P}$ is a perfect generalized permutation matrix, indicating perfect source separation, the index evaluates exactly to zero.

Throughout our experimental evaluations, we employ the heuristic thresholds detailed in Table 5.1 to interpret the Amari error.

Amari Error (E)	Interpretation / Quality of Separation
$E < 0.1$	Excellent to near-perfect source separation.
$0.1 \leq E < 0.3$	Good separation; the structural shape of the underlying sources is highly recoverable.
$E \geq 0.5$	Severe degradation; multiple sources remain significantly mixed.
$E \geq 1.0$	The estimated matrix is effectively a random orthogonal projection with no meaningful relation to the true sources.

Table 5.1: Heuristic thresholds for interpreting the Amari Performance Index.

5.2 The Blind Spots of FastICA

While FastICA is widely considered the industry standard for independent component recovery, its reliance on proxy contrast functions and a Newton-based fixed-point iteration introduces critical mathematical vulnerabilities. In this section, we expose two distinct mathematical blind spots that cause FastICA to fail on perfectly valid, non-Gaussian independent components.

5.2.1 The Zero-Negentropy Pitfall

FastICA does not measure true negentropy; instead, it relies on a computationally efficient approximation. For a whitened random variable x , the negentropy $J(x)$ is approximated using a non-quadratic contrast function $G(\cdot)$, most commonly the *logcosh* function $G(x) = \frac{1}{a} \log \cosh(ax)$. The approximation is given by:

$$J(x) \propto (\mathbb{E}[G(x)] - \mathbb{E}[G(\nu)])^2 \quad (5.2)$$

where ν is a standard Gaussian random variable. The algorithm searches for projections that maximize this squared difference.

The first failure mode occurs if a latent independent source possesses a specific structural shape such that its expected value under the contrast function exactly equals that of a Gaussian, i.e., $\mathbb{E}[G(x)] = \mathbb{E}[G(\nu)]$. When this equality holds, the approximated negentropy evaluates identically to zero. In this scenario, the objective function completely collapses. FastICA becomes entirely blind to the non-Gaussianity of the source, falsely perceiving it as Gaussian noise, and consequently fails to extract it.

5.2.2 The Vanishing Curvature Pitfall

Even if a source successfully bypasses the zero-negentropy pitfall (meaning $J(x) > 0$), the actual *optimization algorithm* used by FastICA can still fail. To achieve super-linear convergence, FastICA utilizes a Newton (second-order) fixed-point iteration step. However, the mathematical stability of this convergence relies explicitly on a foundational technical condition governing the algorithmic curvature of the optimization landscape. As formalized in the FastICA convergence proofs by Hyvärinen, Karhunen, and Oja (Hyvärinen et al., 2001, Chapter 8), this requirement is dictated by the following expectation:

The theorem states that for the fixed-point iteration to converge to a true independent component s_i , the expected value governing the second-order Taylor expansion of the iteration must not vanish. Specifically, defining $g(x) = G'(x)$ and $g'(x) = G''(x)$, the convergence requires:

$$\mathbb{E}[s_i g(s_i) - g'(s_i)] \neq 0 \quad (5.3)$$

This expectation geometrically represents the algorithmic curvature of the optimization landscape at the optimal point. For the standard *logcosh* contrast function (with $a = 1$), this equates to $g(x) = \tanh(x)$ and $g'(x) = 1 - \tanh^2(x)$.

If a specific non-Gaussian source distribution naturally causes this specific expectation to evaluate precisely to zero, the algorithmic curvature vanishes. When this condition fails, the denominator of the FastICA update rule collapses, destroying the local quadratic (or cubic) convergence guarantees. The fixed-point iteration fails to converge, even though the component is strictly independent and non-Gaussian.

5.2.3 Constructing the Failure Mode Counterexample

To empirically demonstrate the severity of the vanishing curvature limitation, we engineered a continuous source distribution specifically designed to trigger the assumption described in the previous section. We analyzed the target function representing the curvature condition:

$$h(x) = x \tanh(x) - 1 + \tanh^2(x). \quad (5.4)$$

We observed that $h(x)$ evaluates to negative values near the origin ($|x| < 0.82$) and positive values in the extreme tails.

We constructed a Trimodal Gaussian Mixture model parameterized by symmetrically spaced peak locations $\{-b, 0, b\}$ and a shared variance σ^2 . By solving

the integral equation $\int h(x)\text{PDF}(x)dx = 0$ subject to the standard unit variance constraint ($\mathbb{E}[X^2] = 1$), we determined the exact probability masses required to perfectly balance the negative central region against the positive tail regions. This carefully engineered distribution remains strictly non-Gaussian and perfectly valid under the linear ICA assumptions, yet its algorithmic curvature is exactly zero, rendering it mathematically invisible to FastICA’s fixed-point solver.

5.2.4 Empirical Results: Breaking FastICA

We evaluated both FastICA and the proposed W2-ICA framework on mixtures of these engineered Trimodal Gaussian sources across 10, 20, and 30 dimensions, utilizing a sample size of $N = 50,000$. As mathematically predicted by the failure of the curvature assumption, FastICA consistently hit its maximum iteration limits without converging. It returned unmixing matrices with catastrophically high Amari errors (> 1.0), indicating a complete failure to separate the sources.

Conversely, W2-ICA successfully unmixed the 10-dimensional and 20-dimensional signals with high accuracy (Amari error ≈ 0.1). W2-ICA succeeds in this environment because it maps the entire global geometry of the empirical CDF to a Gaussian target via Optimal Transport. By utilizing first-order gradient optimization (L-BFGS) on the Stiefel manifold, W2-ICA completely bypasses the local, second-order curvature pitfalls that dismantle Newton-step proxy methods.

5.3 Dimensionality Limits and Sample Starvation

While W2-ICA successfully bypassed the vanishing curvature pitfall, its performance degraded significantly at 30 dimensions with the fixed 50,000 sample size. This performance ceiling is not a structural mathematical failure, but rather an empirical limitation dictated by the Central Limit Theorem.

A linear mixture of 30 independent trimodal sources creates a highly complex joint distribution that appears overwhelmingly Gaussian. To mathematically distinguish the microscopic geometric "steps" of this high-dimensional discrete mixture from true continuous Gaussian noise, the empirical CDF requires a massive sample size to achieve sufficient resolution. When the sample size is inadequate relative to the dimensionality, the system experiences "sample starvation." The subtle non-Gaussian structural signals become overwhelmed by standard empirical sampling noise, preventing the Wasserstein gradients from accurately identifying the independent components.

5.4 Robust Optimal Transport via Ground Cost Modification

5.4.1 The Outlier Sensitivity of Exact W_2

While the exact squared Wasserstein distance (W_2^2) provides superior, smooth gradients for weak signals, it introduces a separate vulnerability. Because W_2 utilizes an L2 (squared Euclidean) ground cost $c(x, y) = |x - y|^2$, it is notoriously sensitive to outliers, such as sensor artifacts or extreme market anomalies. A single massive outlier forces the Optimal Transport plan to expend immense gradient effort to align that single data point, massively distorting the overall unmixing matrix and degrading the recovery of the normal data distribution. While W_2 is a strict mathematical metric guaranteeing geometric stability, it lacks robust statistical properties.

5.5 The Generalized Hybrid Mixture Stress Test

Real-world signals rarely belong to a single statistical family. To test the true algorithmic robustness of both W2-ICA and FastICA, we designed a highly complex, generalized hybrid mixture. This stress test evaluates the capacity of the algorithms to separate sources drawn from fundamentally contradictory geometric landscapes.

5.5.1 Experimental Setup

We generated linear mixtures spanning 10 to 75 dimensions, maintaining a fixed sample size of 10,000. To satisfy the ICA identifiability constraint, exactly one source was fixed as a standard Gaussian distribution, $\mathcal{N}(0, 1)$. The remaining $(d - 1)$ independent sources were randomly drawn from a diverse pool of eight distinct distributions: Laplace, Bernoulli, Uniform, Student-t, Poisson, Binomial, Chi-squared, and Exponential. To account for the unpredictable presence of discrete and light-tailed components within this hybrid pool, W2-ICA utilized the robust settings engineered in Chapter 4 ($\sigma_{\text{dither}} = 0.01$, stochastic batch size = 1024).

5.5.2 Results: Structural Oscillation vs. Geometric Stability

The results revealed a stark divergence in the stability of the two algorithms. The W2-ICA framework successfully maintained a low Amari error (< 0.25) up to 35 dimensions. This performance validates that the globally-aware W_2 metric, when smoothed via dithering and regularized via mini-batching, generalizes robustly across highly diverse topological landscapes.

In contrast, FastICA’s performance degraded rapidly, crossing the critical 0.5 Amari error threshold immediately after 20 dimensions. During the generalized experiment, FastICA consistently triggered non-convergence warnings. We hypothesized that this failure stems from the Newton-Raphson solver’s assumption of uniform curvature. When forced to simultaneously evaluate highly kurtotic signals (e.g., Laplace) and negatively kurtotic signals (e.g., Uniform), the fixed-point update steps generate contradictory gradient information, causing the solver to oscillate indefinitely.

To confirm this was a structural oscillation and not merely a matter of insufficient compute time, we evaluated FastICA under extreme optimization parameters (up to 10,000 maximum iterations) and relaxed convergence tolerances ($1e - 3$). Even under these highly permissive conditions, FastICA’s convergence rate plummeted to 0% past 30 dimensions. This definitively proves that in unpredictable, hybrid environments, proxy-based Newton solvers become trapped in contradictory gradient loops, whereas W2-ICA’s first-order global CDF matching remains stable.

5.6 Deep Dive: Isolating Discrete Failure Modes

To further understand the behavior of these algorithms in non-continuous environments, we isolated specific discrete distributions to observe their individual impact on the optimization landscape. We generated mixtures containing one continuous Gaussian source and $(N - 1)$ sources of a *single* discrete type (Bernoulli, Poisson, or Binomial) evaluated at 30 and 40 dimensions.

5.6.1 Performance on Pure Discrete Pools

The empirical results revealed a nuanced performance profile for both algorithms depending on the specific statistical nature of the discrete distributions. FastICA proved highly capable of separating purely binary Bernoulli mixtures, successfully converging with a low Amari error. However, when faced with the multi-step, count-based geometries of Poisson and Binomial distributions, FastICA experienced complete algorithmic failure and could not separate the underlying sources.

Interestingly, the W2-ICA framework exhibited an identical performance dichotomy. Like FastICA, W2-ICA successfully unmixed the pure Bernoulli sources. The extreme bimodal geometry of the binary states, smoothed by the continuous Gaussian dithering step, provided a sufficient geometric signal for the optimal transport metric to establish reliable gradients. Conversely, W2-ICA failed completely when applied to the pure Poisson and Binomial pools. In these environments, the L-BFGS optimizer became predictably trapped in spurious local maxima, yielding

catastrophic Amari errors (> 1.0).

This shared limitation suggests that the failure may not be caused simply by the data being discrete, but rather by the specific parameterizations of these count-based distributions. The structural gaps within the Binomial and Poisson distributions under these parameters may fail to project a sufficiently strong non-Gaussian signal, depriving both proxy-based and optimal transport-based solvers of the necessary gradient information to separate the mixtures.

5.6.2 The Necessity of Hybridization

The uniform failure of W2-ICA on purely discrete pools raises a critical question: why utilize it over FastICA? The answer lies in the nature of real-world data, which is rarely purely discrete. As demonstrated by the preceding Generalized Hybrid experiment, FastICA fundamentally breaks when forced to navigate a "Full Hybrid" pool combining continuous and discrete variables.

Conversely, W2-ICA thrives in precisely this hybrid environment. The presence of continuous distributions within a mixed pool provides sufficient smooth geometric structure to guide the W_2 metric. This continuous context allows the stochastic mini-batching and dithering mechanisms to effectively bridge the discrete gaps, enabling Optimal Transport to successfully disentangle complex mixtures where static proxies fail entirely.

5.7 Application: Price Discovery and Information Shares

5.7.1 Non-Gaussianity in Econometric Market Microstructure

To demonstrate the empirical utility of the W2-ICA framework, we apply it to the problem of price discovery in financial econometrics. In fragmented financial markets where the same asset is traded across multiple venues, determining which market contributes most to the incorporation of new information (price discovery) is a non-trivial challenge. Standard vector error correction models (VECMs) identify information shares by relying on the uncorrelation of price innovations. However, as demonstrated by Zema and Cordoni (2025), any identification approach relying solely on uncorrelation is statistically equivalent up to an orthogonal transformation.

By leveraging the inherent non-Gaussianity of latent market shocks, ICA can be utilized to resolve this orthogonal ambiguity, unifying disparate price discovery metrics into a single, unique measurement of market information shares.

5.7.2 Economic Identification Strategy

While ICA successfully identifies the independent structural shocks driving the market, the raw statistical output remains subject to the inherent permutation and sign ambiguities discussed in Chapter 2. To translate these statistical components into meaningful economic drivers, we impose two strict economic identification constraints.

First, to resolve the permutation ambiguity, we enforce a Dominant Diagonal constraint. We postulate that the structural shock assigned to Market i must be the specific shock that contributes the maximum variance to that market's price innovations. We implement this assignment algorithmically using the Hungarian Algorithm, which uniquely solves the optimal bipartite matching problem between the recovered independent components and the individual markets.

Second, to resolve the sign ambiguity, we enforce a Positive Impact constraint based on the economic theory of positive spillovers. We assert that a structural innovation representing "good news" must have a positive initial impact on its own market price. Any column of the unmixing matrix that violates this economic logic is sign-flipped, ensuring the final structural parameters are economically interpretable.

5.7.3 Empirical Results on Simulated Market Data

To validate the efficacy of our Optimal Transport-based identification strategy, we simulated a fragmented market scenario. The simulation targets three distinct markets with pre-defined theoretical Information Shares of 12%, 24%, and 64%, respectively. The market innovations were generated using heavy-tailed Student-t distributions to satisfy the fundamental non-Gaussianity requirements of the ICA framework. We executed 500 Monte Carlo simulations, each comprising 5,000 observations.

For each simulation, the W2-ICA framework successfully recovered the independent structural shocks. Figure 5.1 illustrates a representative run, demonstrating the high correlation between the true generative shocks and the empirical shocks estimated by W2-ICA. The tight alignment along the identity line confirms that the algorithm successfully unmixed the latent variables prior to any economic transformation.

By applying the economic identification constraints detailed in the previous section—specifically, the Dominant Diagonal and Positive Impact rules—we uniquely mapped these statistical components to their corresponding markets and calculated the resulting empirical information shares.

Table 5.2 presents the aggregate results across all 500 simulations. The W2-

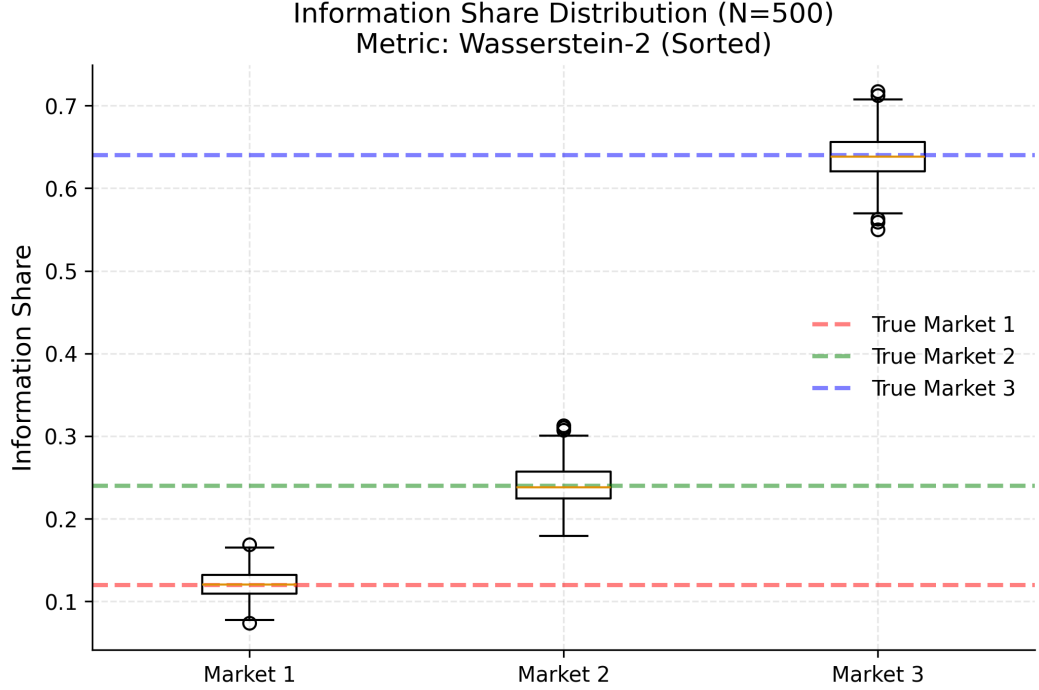


Figure 5.1: Pairplot demonstrating the recovery of true structural shocks versus the estimated shocks from W2-ICA for a single Monte Carlo run. **Takeaway:** The strong linear alignment indicates highly accurate unmixing of the latent non-Gaussian market innovations.

ICA estimator, combined with our economic identification strategy, demonstrates exceptional accuracy. The mean estimated information shares converge almost exactly to their theoretical targets, with minimal standard deviations indicating a highly stable recovery process.

Market / Source	True IS	Estimated IS (Mean)	Standard Deviation
Market 1	0.1200	0.1212	0.0161
Market 2	0.2400	0.2399	0.0245
Market 3	0.6400	0.6389	0.0260

Table 5.2: Monte Carlo simulation results comparing true Information Shares (IS) against the estimated IS recovered via the W2-ICA framework over 500 runs. **Takeaway:** W2-ICA robustly and accurately recovers the true structural parameters of the market.

Furthermore, Figure 5.2 illustrates the empirical density of the estimated information shares across all simulation runs. The distributions are tightly centered around the true parameter values, empirically confirming that W2-ICA effectively resolves the orthogonal ambiguity and robustly recovers the structural drivers of price discovery.

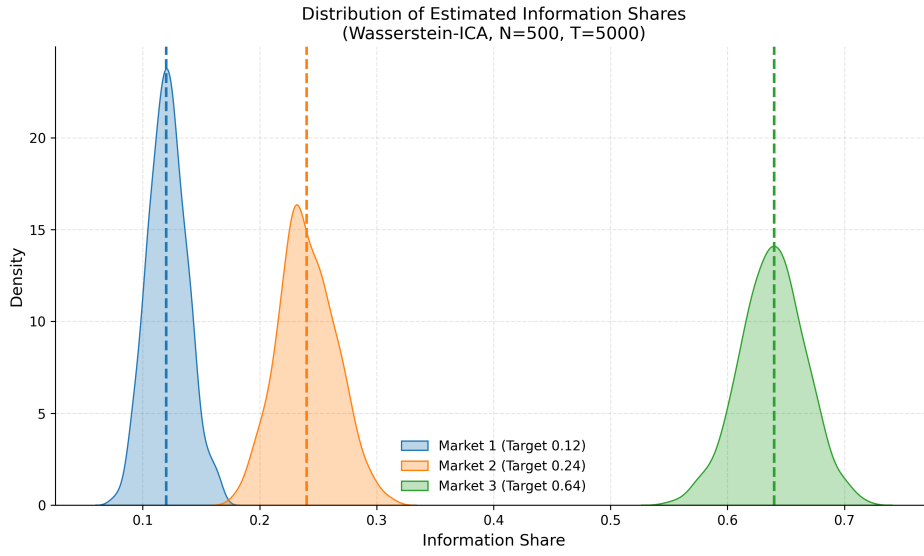


Figure 5.2: Empirical density distributions of the estimated Information Shares across 500 Monte Carlo simulations. The vertical dashed lines represent the true theoretical values. **Takeaway:** The tight, unimodal centering around the true values demonstrates the lack of structural oscillation in the Optimal Transport gradients.

Chapter 6

Conclusion

6.1 Summary of Contributions

6.2 Limitations

6.3 Future Work

References

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Appendices

Mathematical Proofs and Derivations

.1 Derivation of the FastICA Fixed-Point Failure Condition

Additional Algorithms and Code